

Bioinformatics for engineers

Sequencing methods

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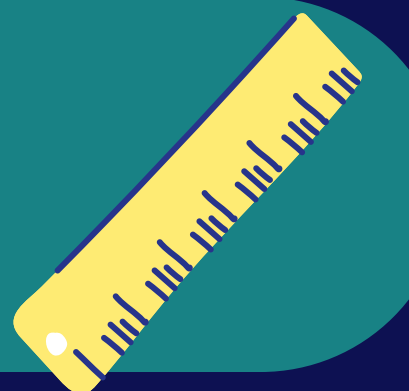
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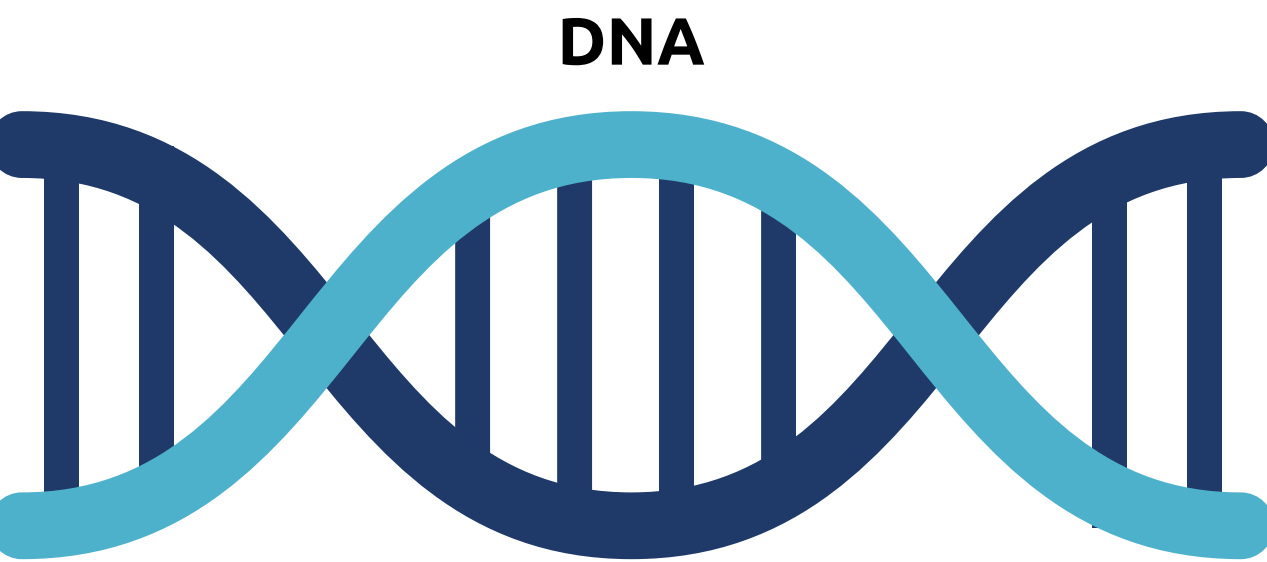
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THE CENTRAL DOGMA



DNA

CENTRAL DOGMA

a biological decompression algorithm

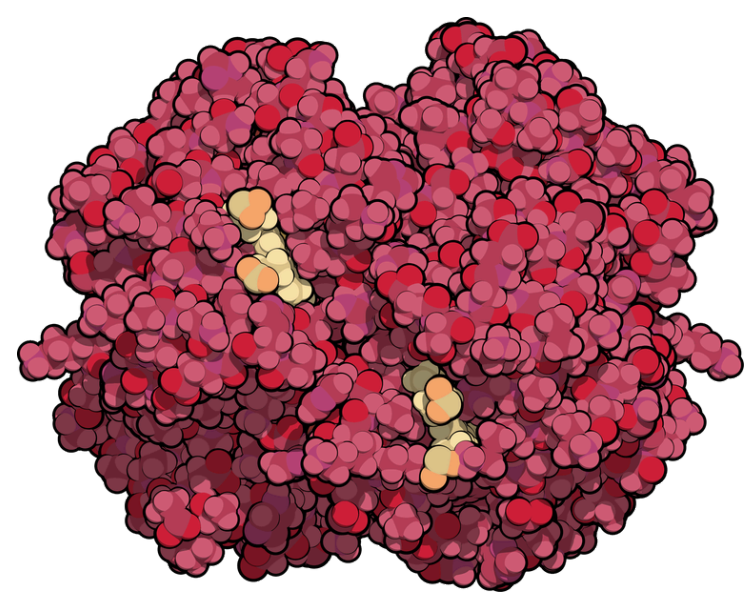
Gene

3' A C A C T T A T ... A G G 5'

3' T G T G A A

RNA_m

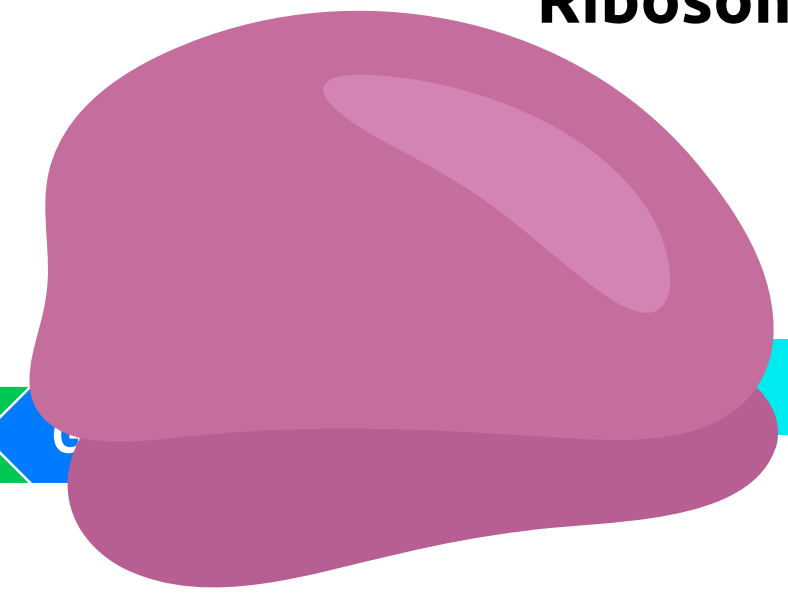
U A .. U C C



Protein



Ribosome



RNA_m

U U C C

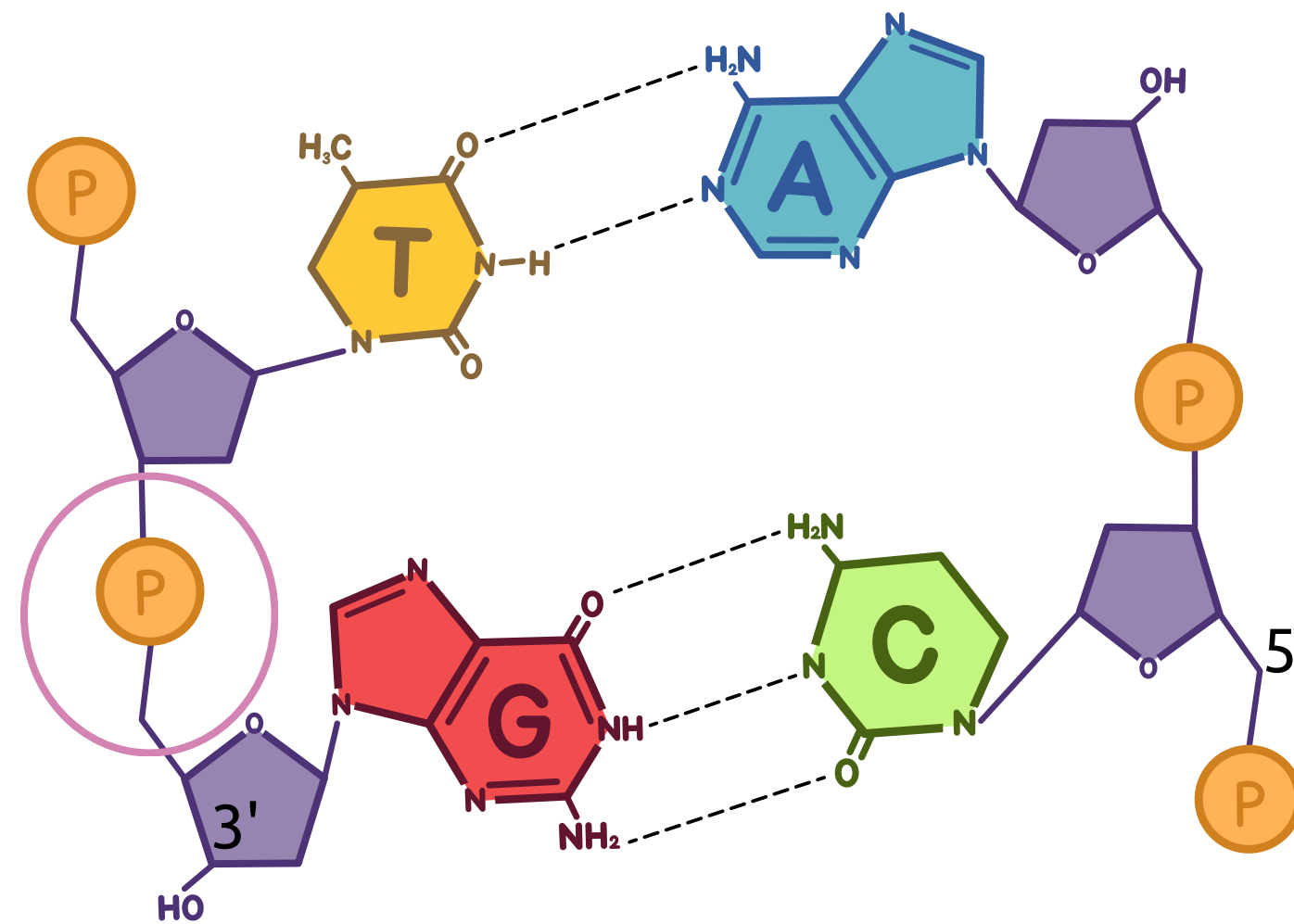
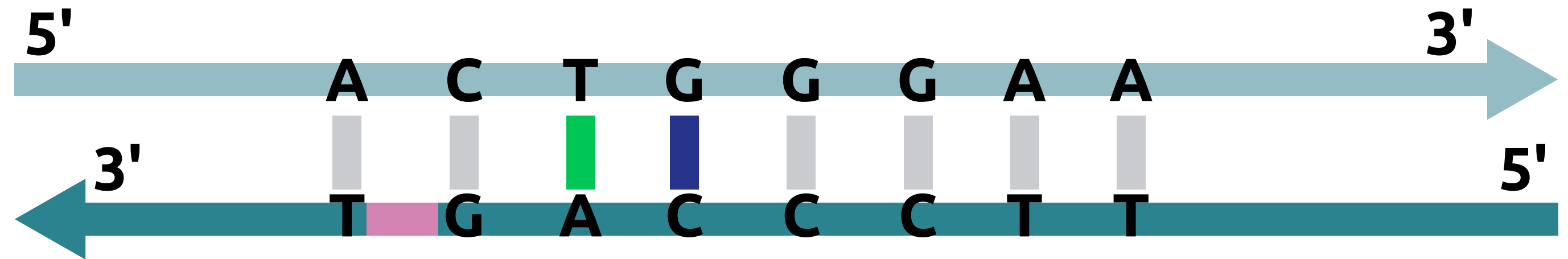
F L M T A L C

AA Seq

THE CHEMISTRY OF THE NUCLEOTIDES



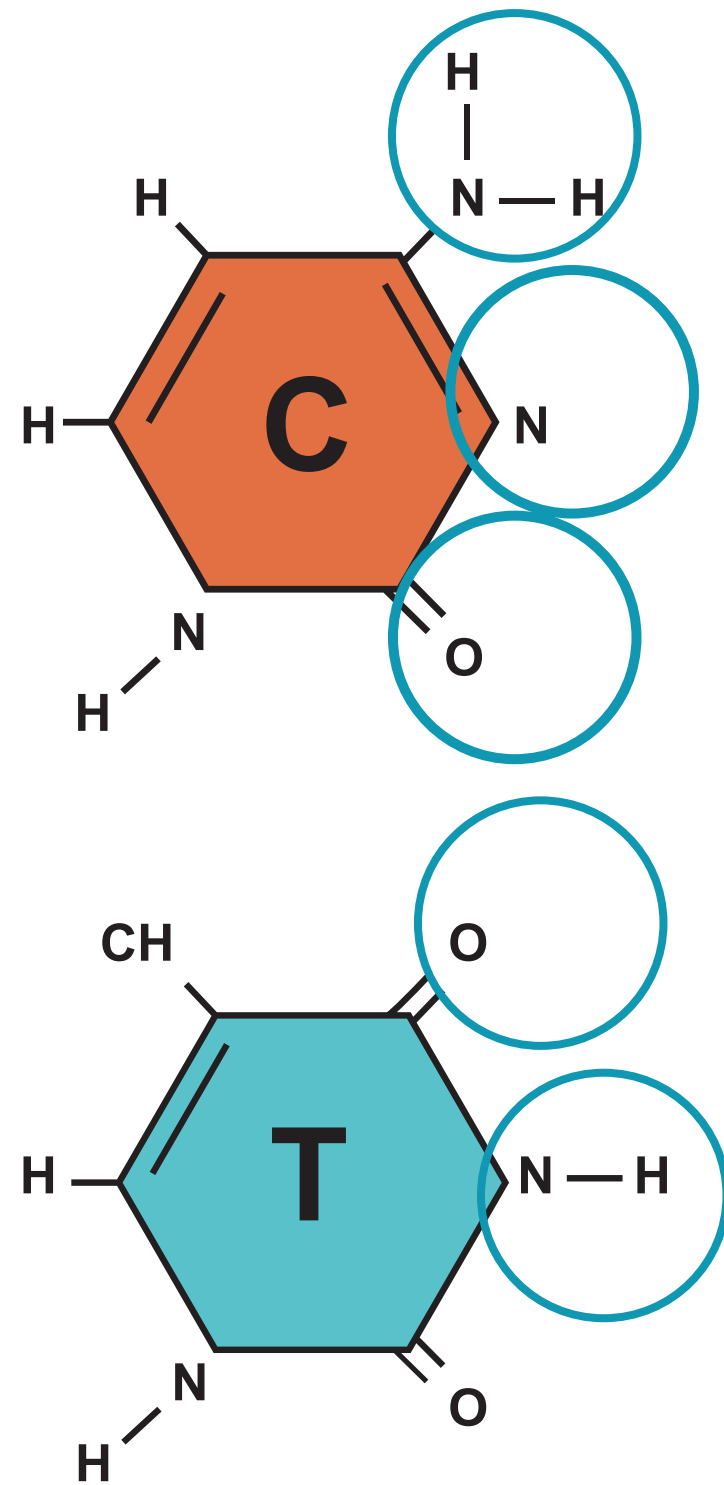
The macro sturture



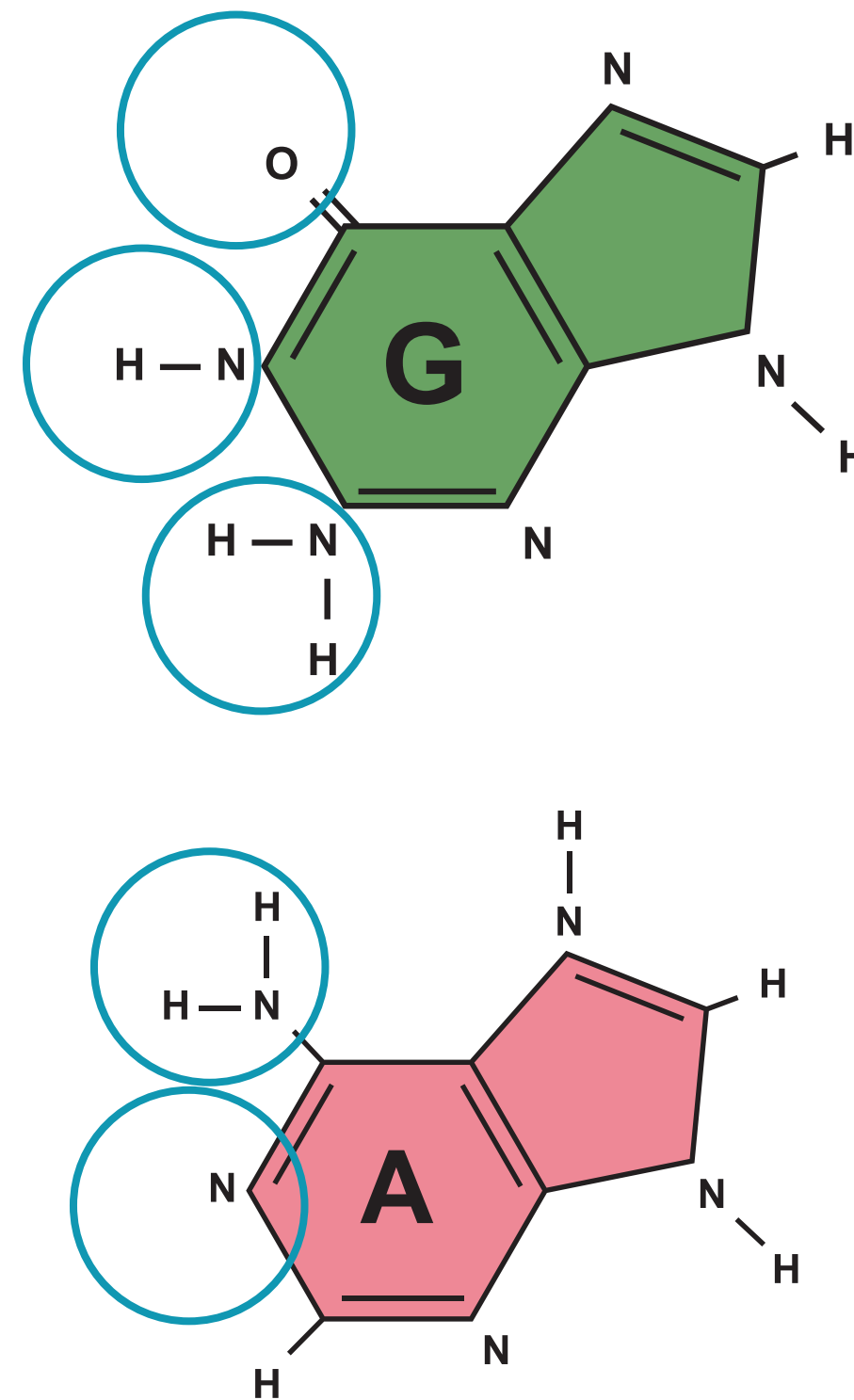
- The DNA structure is a double-helix.
- Hydrogen bridges join the nucleotides of the second chain.
- phosphodiester bond, joins the nucleotides of the same chain.
- The number of hydrogen acceptors mediates pairing specificity.

The micro structure

Pyrimidines

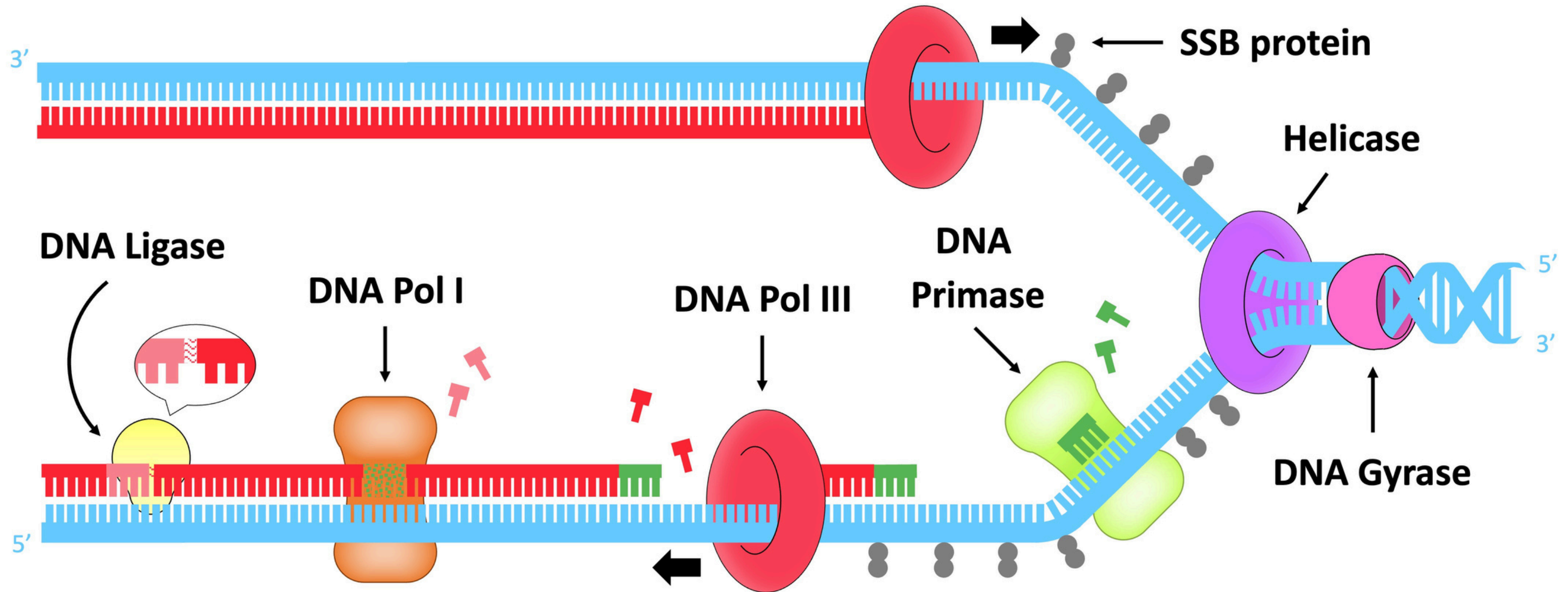


Purines



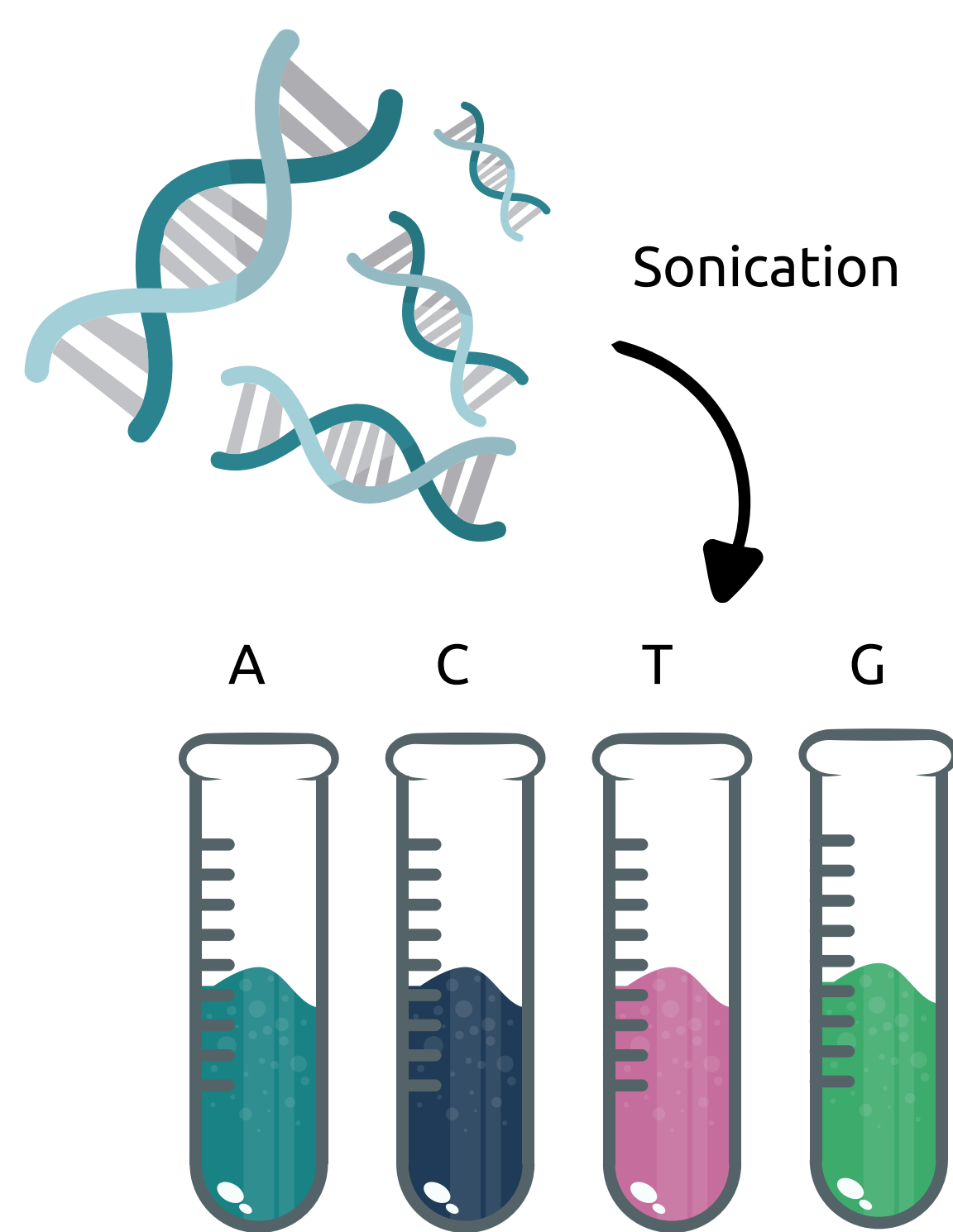
POLYMERASE CHAIN REACTION

The polymerase action

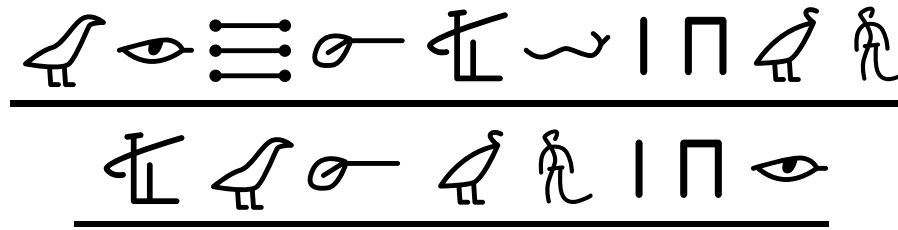


<https://www.youtube.com/watch?v=SMLSAl5igeY>

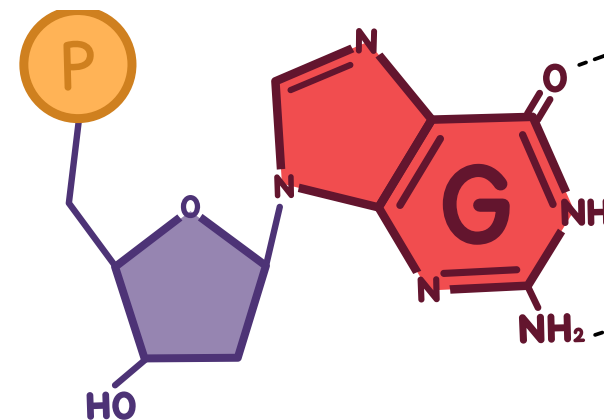
SANGER METHOD



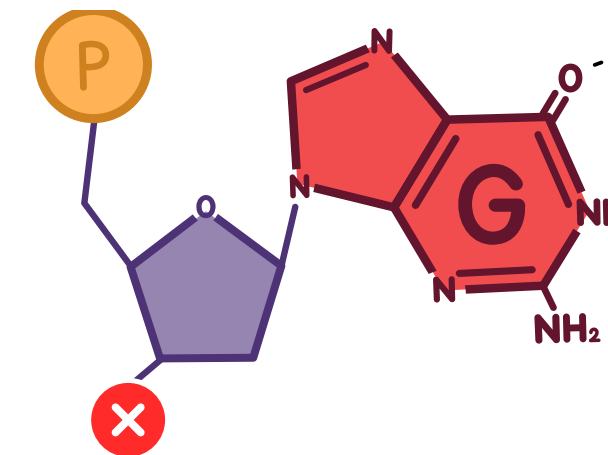
The ancient way



dNTPS



ddNTPS



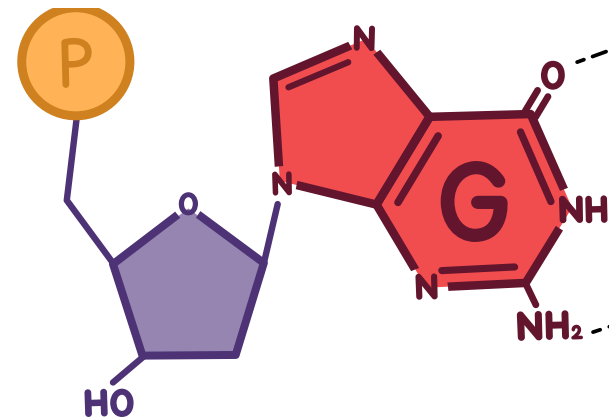
- ddNTPS stops the reaction.
- The weight of the chains is associated with their length.
- Doing the process for each nucleotide allows the reconstruction of the chain.
- Plasmids are used for whole genome sequencing.
- from 500 up to 1000 pb.

An improvement

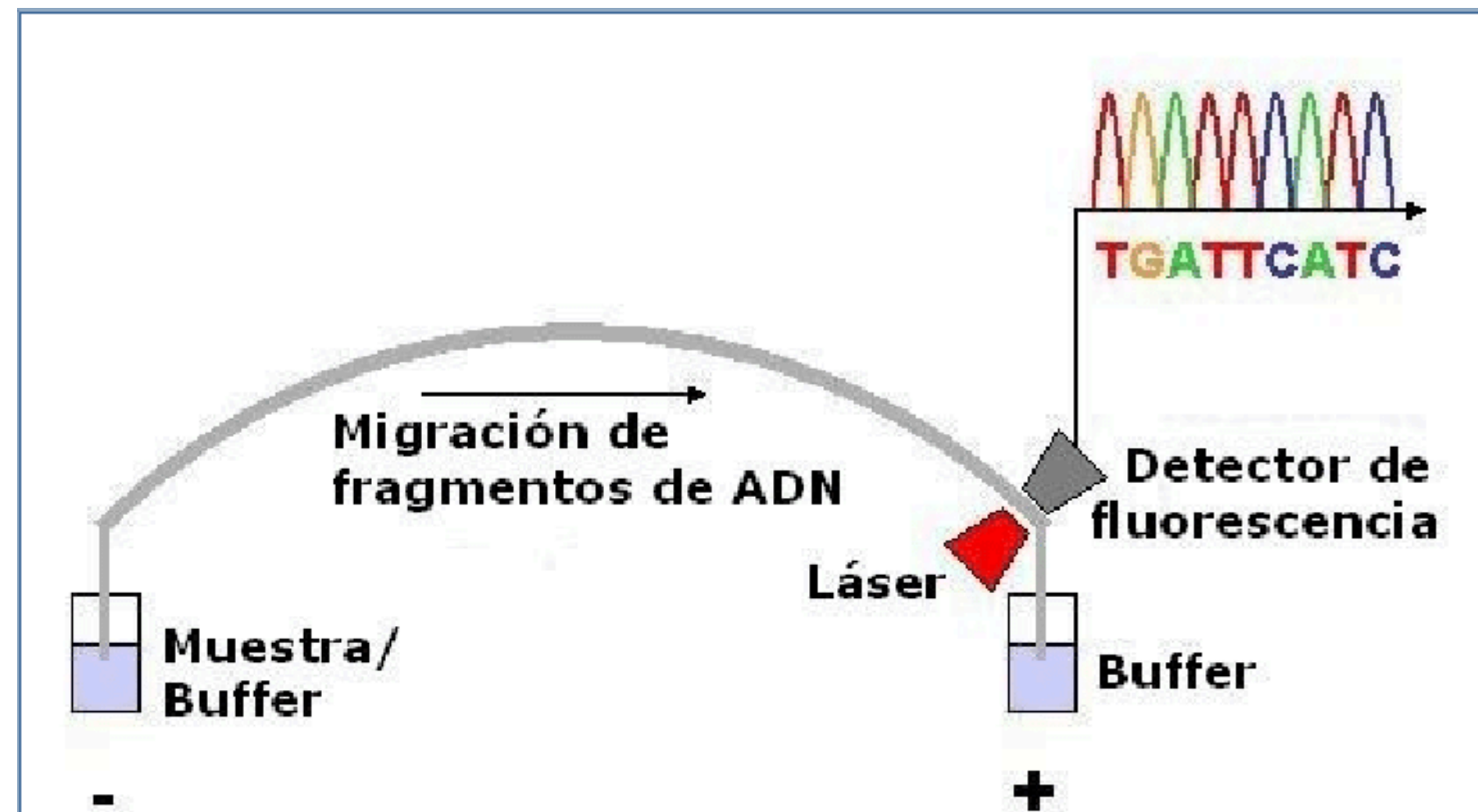
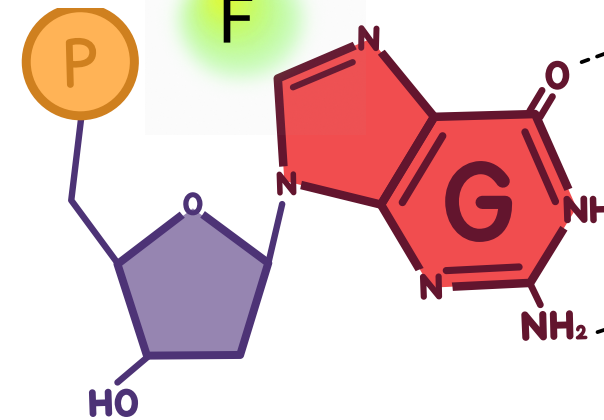
Sonication

A, C, T, G

dNTPS



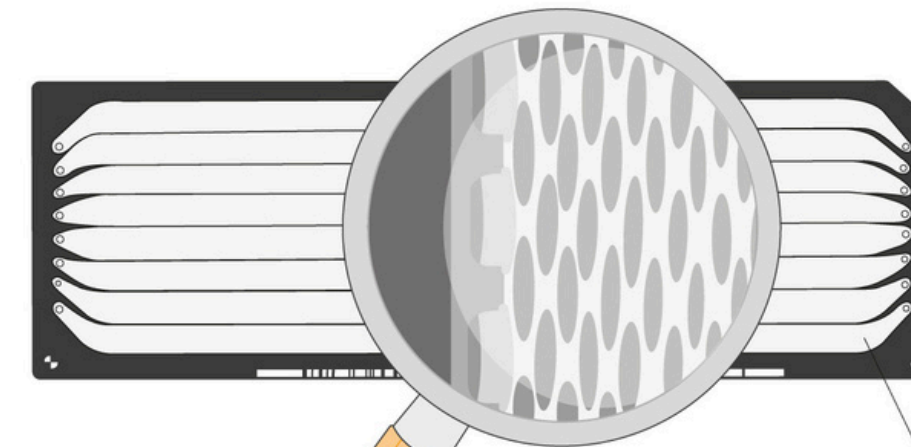
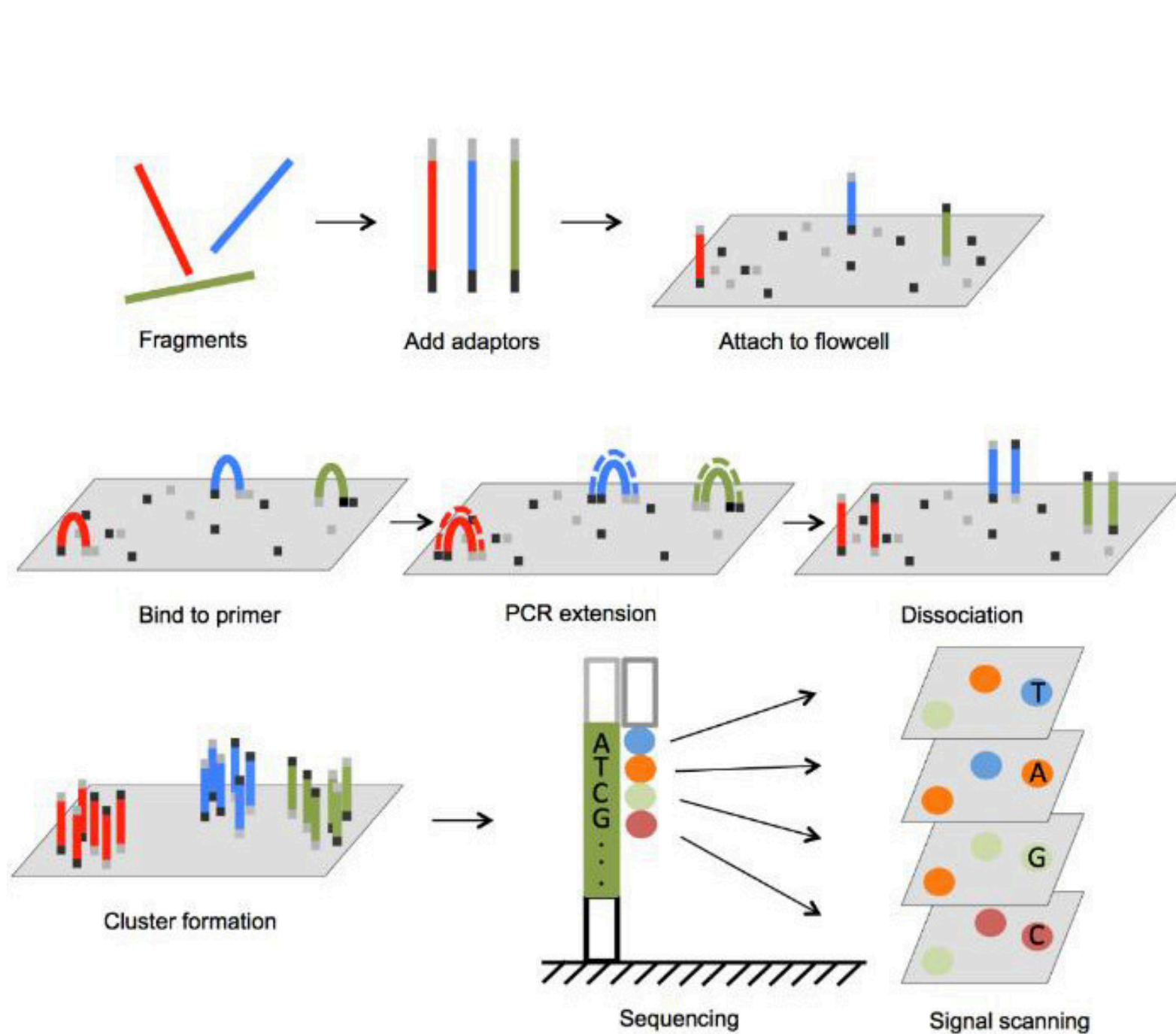
ddNTPS



ILLUMINA

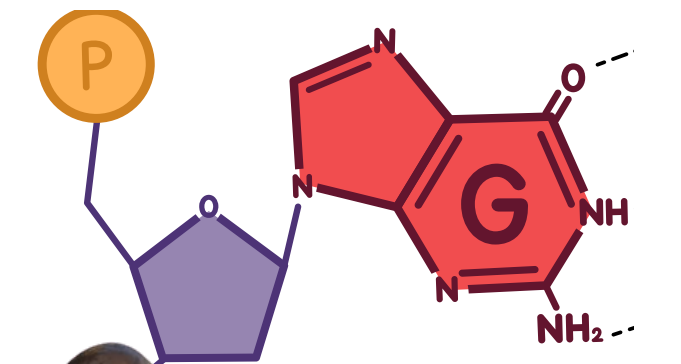
NGS

Single end



- After the cluster formation only R1 is read (5' to 3' sequence) , getting as a results a 150 pb read.
- O-azidomethyl could be transformed to OH using sodium sulfite.

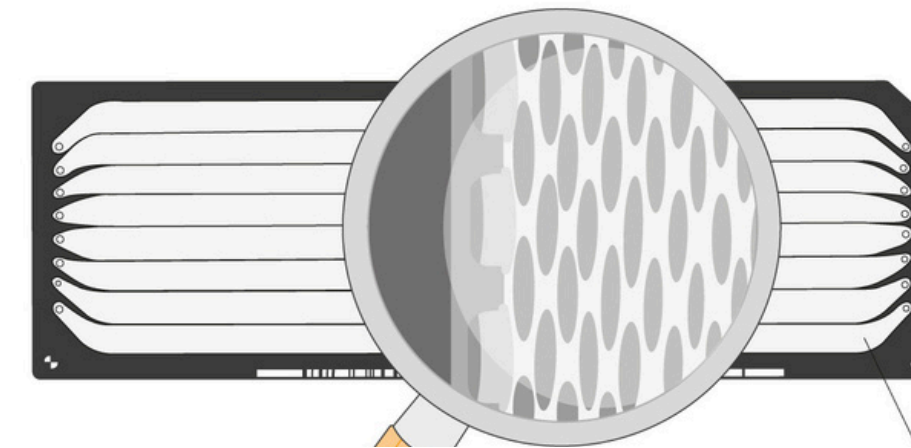
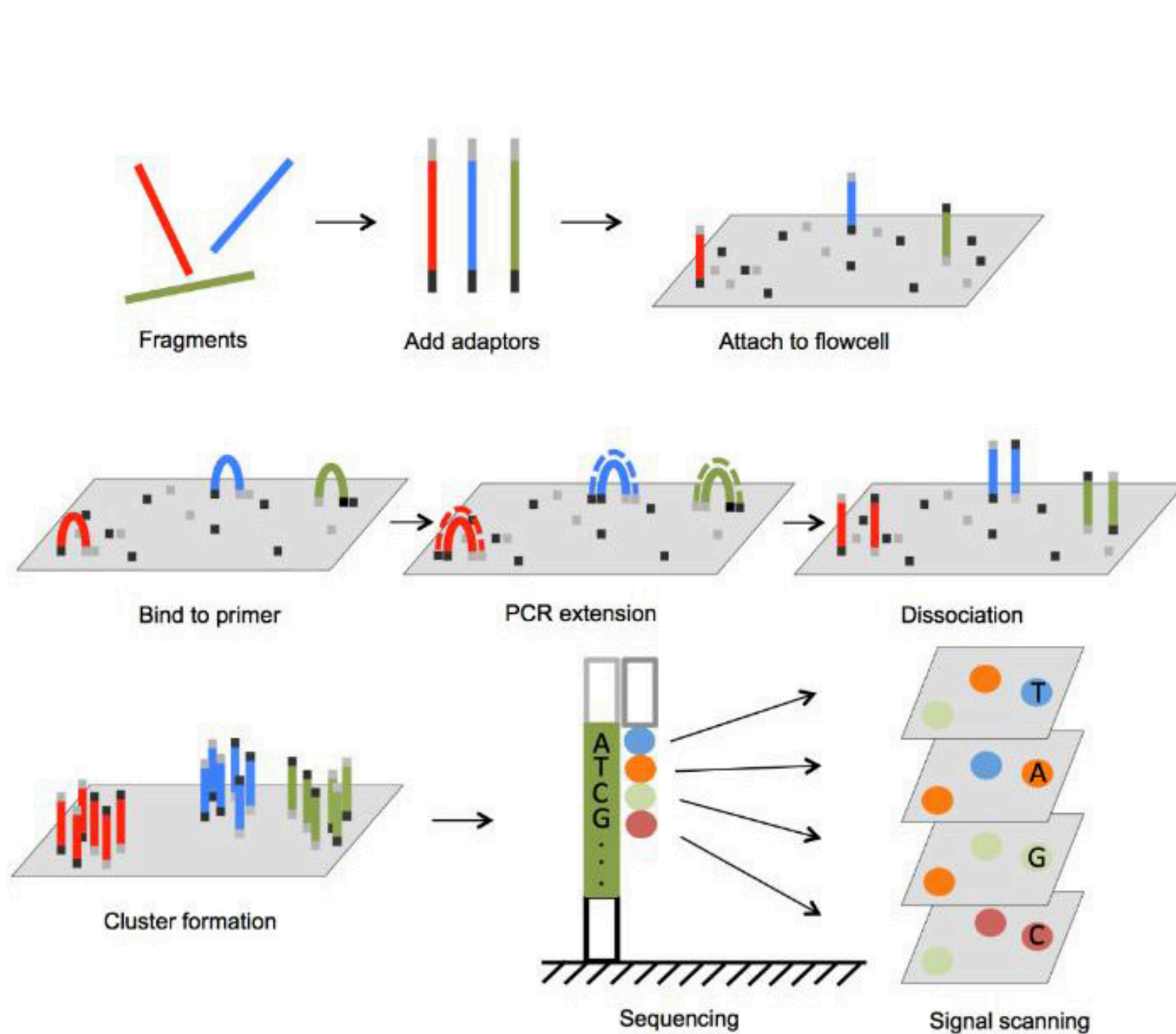
Block-dNTPS



O-azidomethyl

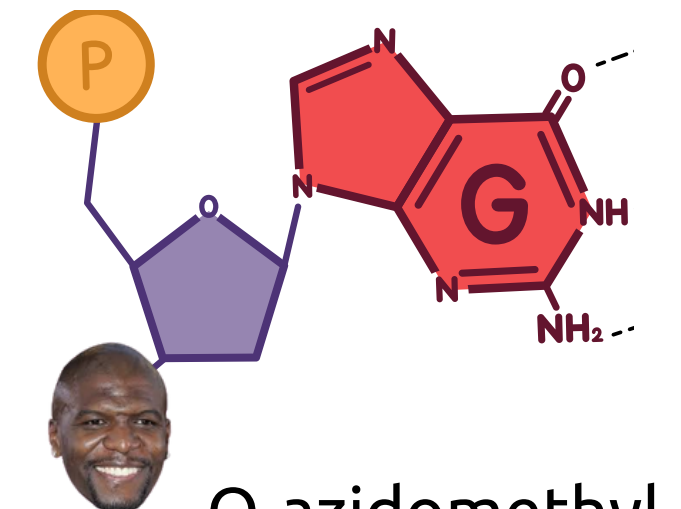
NGS

Paired end



- After the cluster formation, R1 and R2 are reads resulting in longer reads, up to 300 pbs.
- A second option is that R1 and R2 are not contiguous. Allowing the obtention of two reads with a clear distance between both.

Block-dNTPS



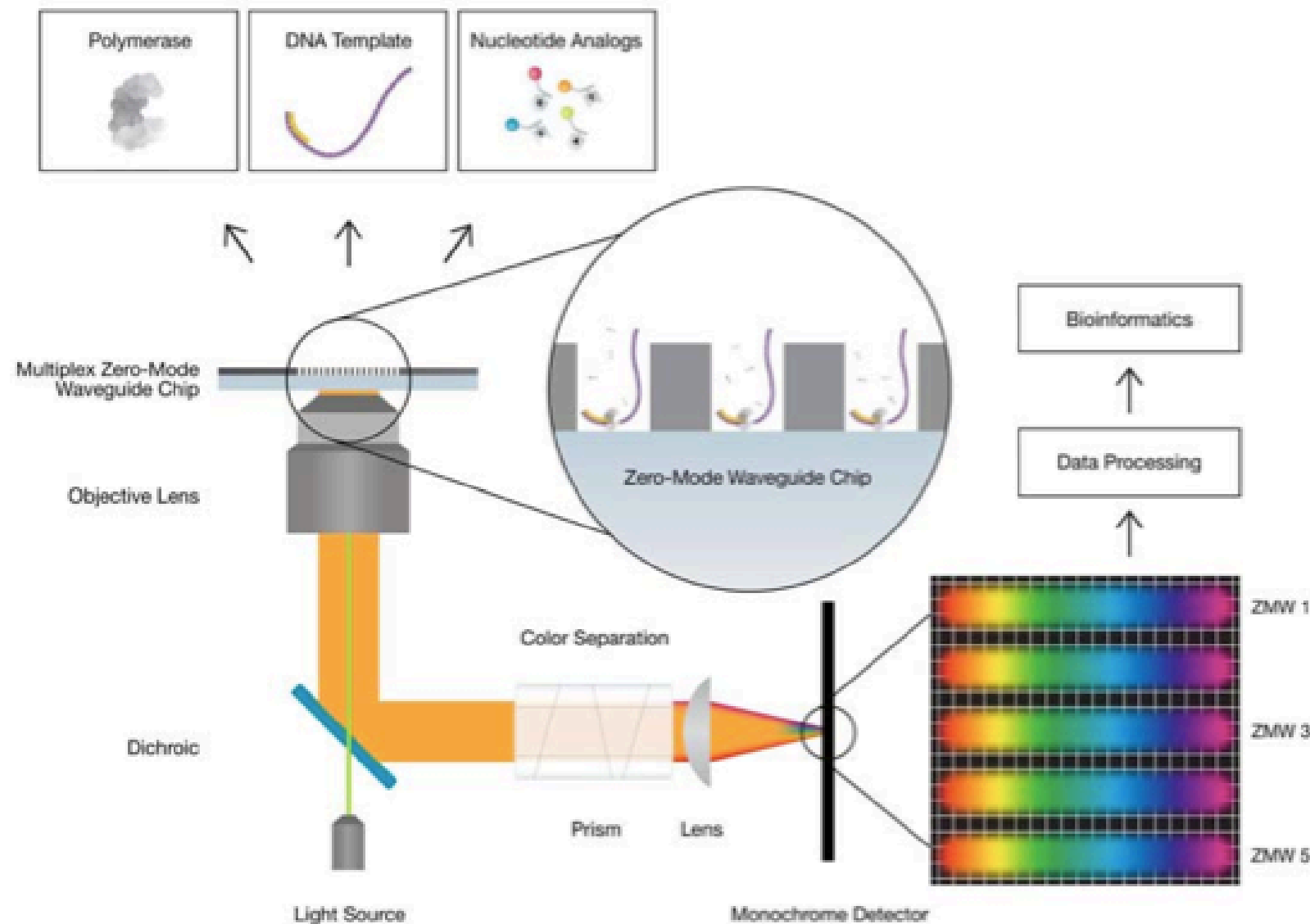
O-azidomethyl

PACBIO



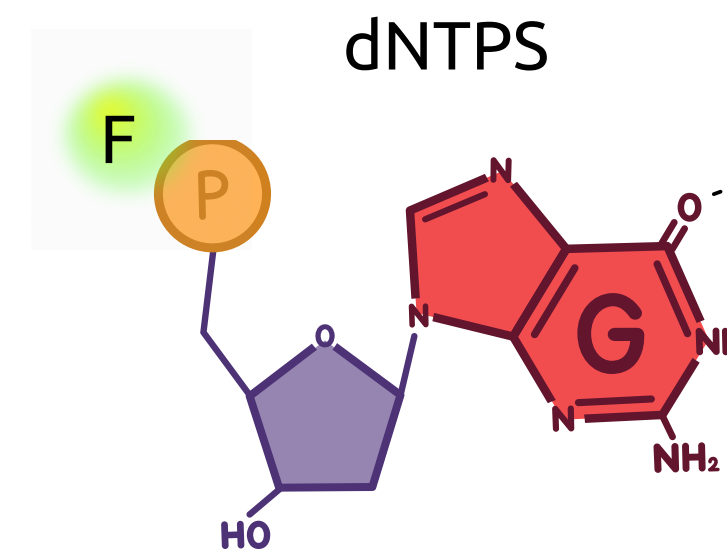
SMART

Single Molecule, Real-Time



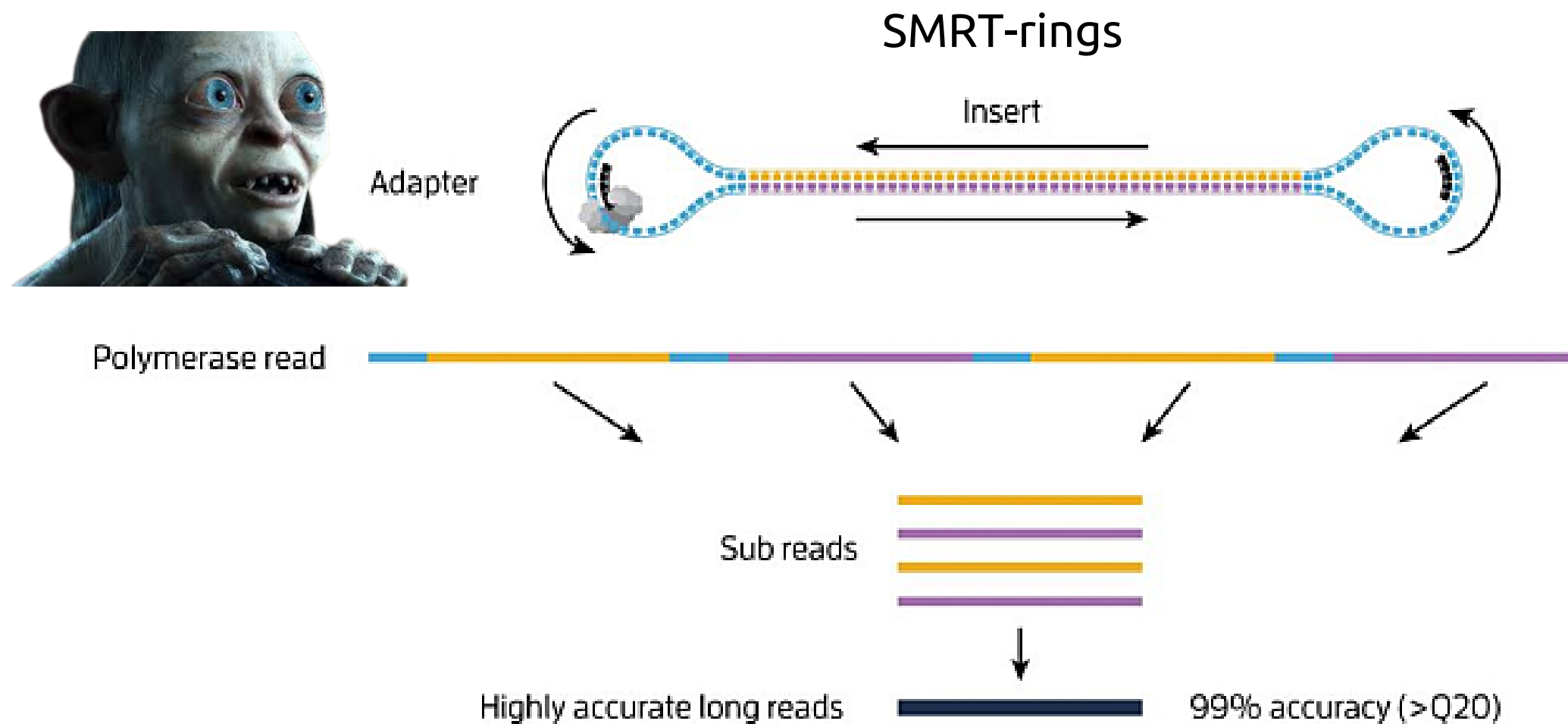
Zero-Mode Waveguides

- PacBio uses a plate of nanopores, where each one contains a static polymerase on its bottom.
- During the polymerase reaction, only the bottom of the deposits are excited by a guided laser, allowing measurement of the nucleotide that it's being added to the chain.
- Each polymerase reaction took a time inside 1 to 10 ms.



SMART

Single Molecule, Real-Time

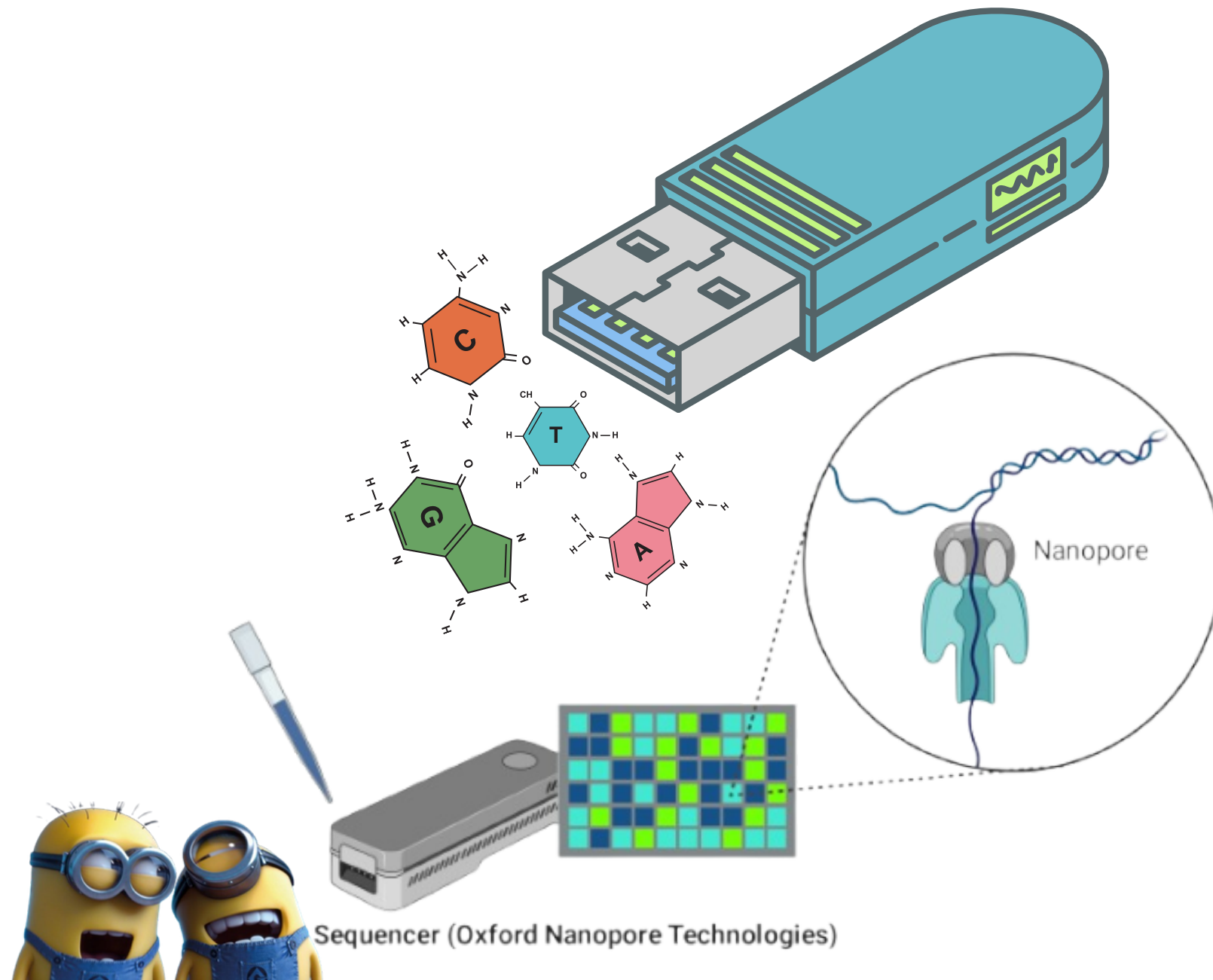


- PacBio uses Phi 29 polymerase. This enzyme can also work as a helicase.
- The topological circle form allows an infinite reading of the original molecule.
- Repeating this process thousands of times, it is possible to extract a consensus to avoid error.
- The obtained reads are in the range of 10 to 100 Kb.

OXFORD NANOPORE

The USB of sequencing

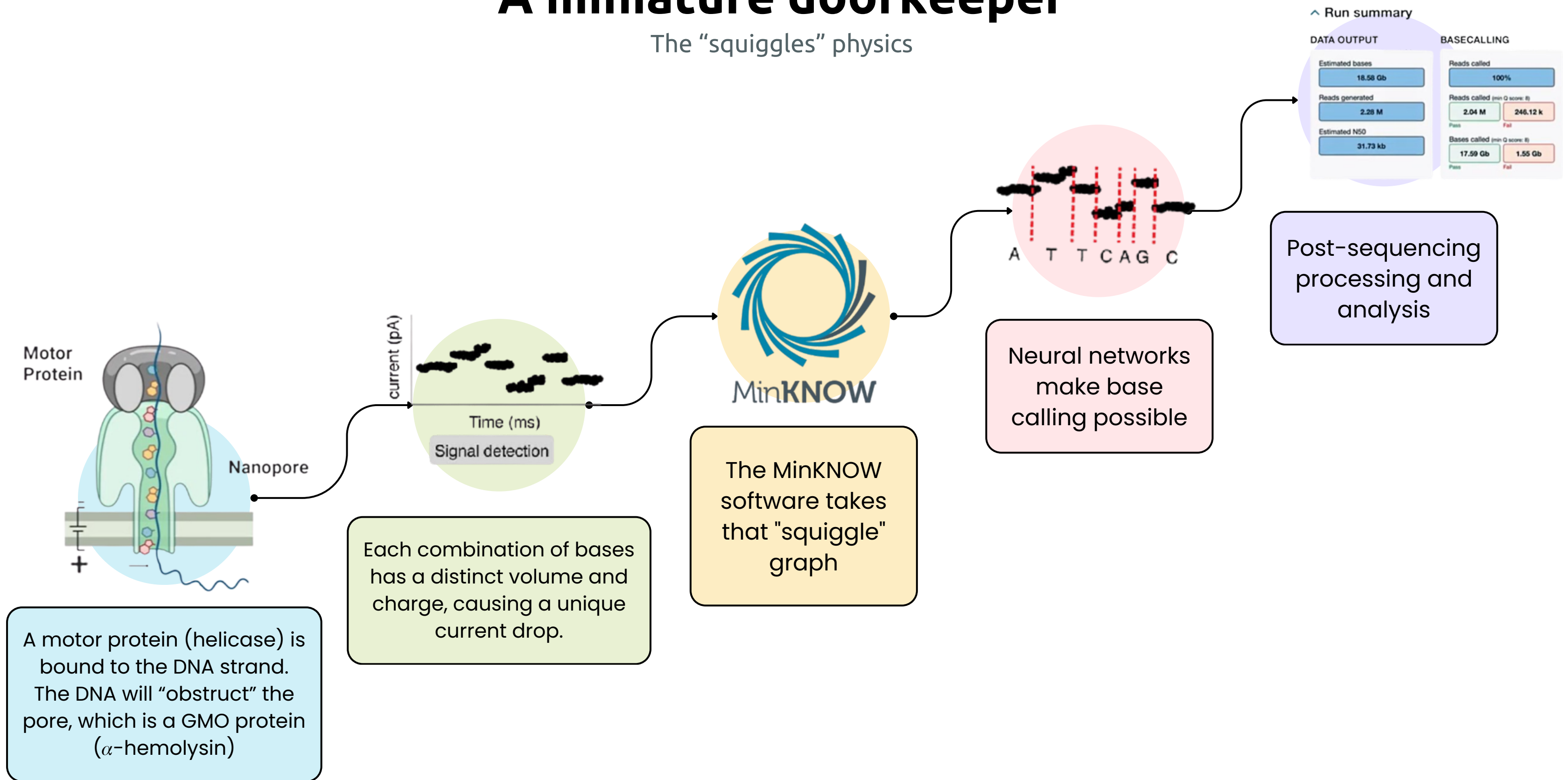
How to apply physics in sequencing



- Imagine you're a bouncer at a bar, but now you don't need to - see- **who is going through the door** (no fluorescence).
- Simply put, each person (nucleotide) is a **different size** and **blocks the entrance differently** as they pass through.
- **So what makes it special?** Take nanopore as a movement sensor, it saves energy processing compared to other methods. Now we are talking of **Ultra-long reads** that you can watch in real time, as the DNA passes through the pore, the data appears on your screen.
- This method is "**PCR-free**", In almost all other methods, DNA is used as a "template" to create a new copy. In Nanopore, ***you read the original molecule.***

A miniature doorkeeper

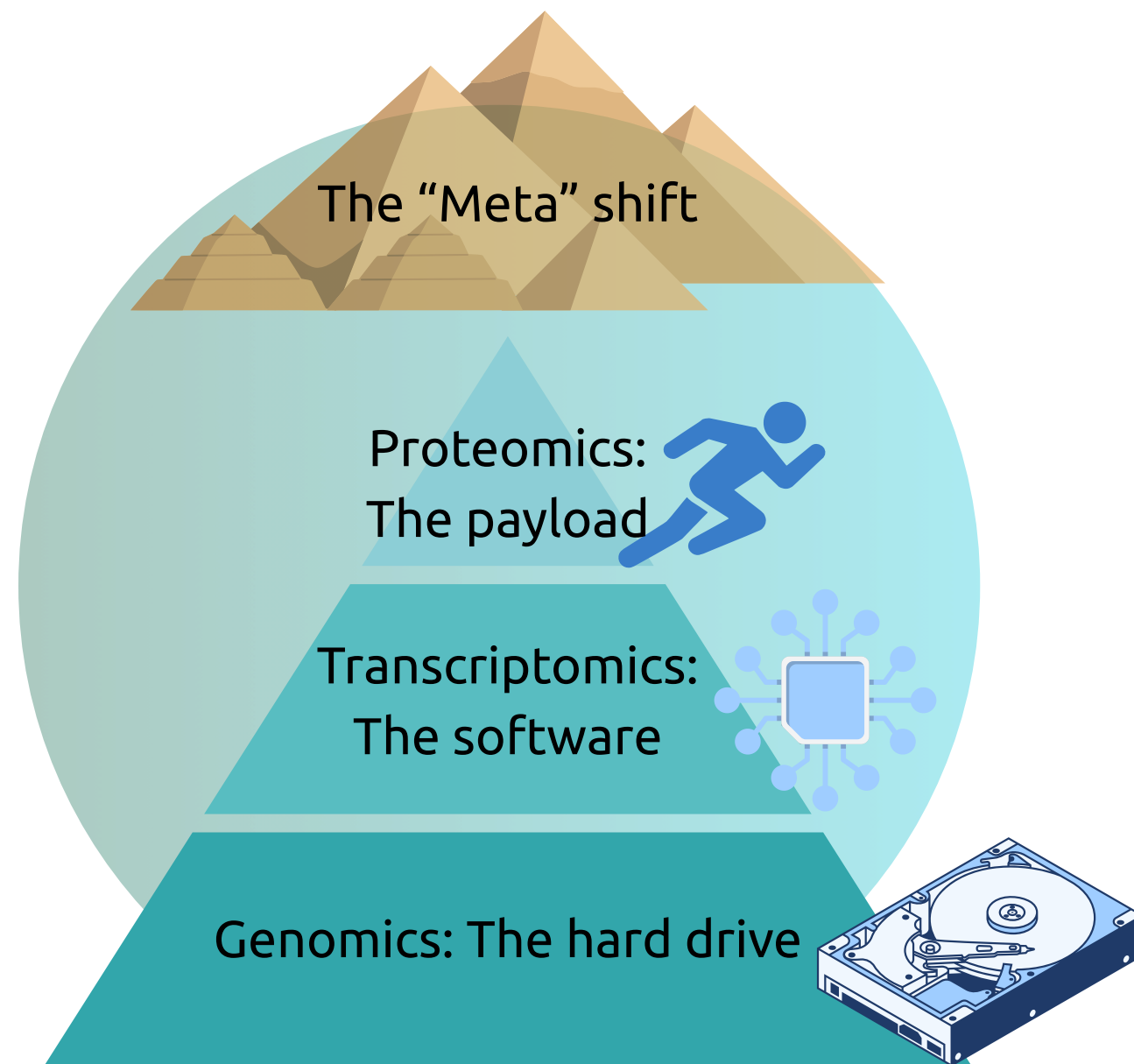
The “squiggles” physics



THE OMICS WORLD

A biological tech stack

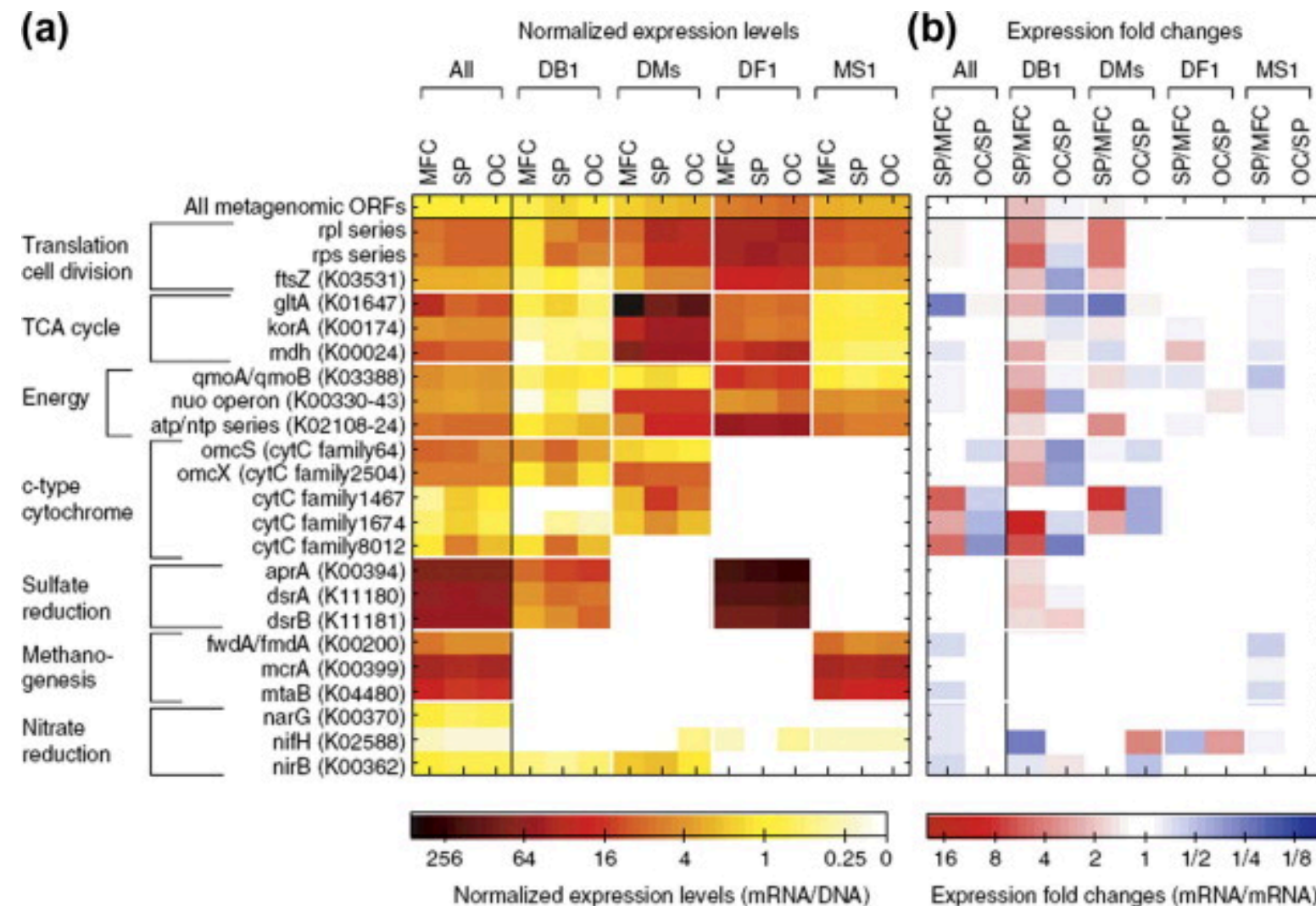
¿What do we mean by “OMICS”?



- The suffix “omics” comes from -oma, which means entirety. For example, genome = all the genes of an **organism**.
- Notice that we represent the omics with a vertical view, now think of **metagenomics** as a “**horizontal view**”.
- In **metagenomics** we study all genetic material (DNA) recovered directly from environmental samples. Now we won’t study one single pyramid, but **THOUSANDS**.
- The goal is to answer “**Who is there?**” and “**What are they capable of doing?**”.

Metatranscriptomics

The "Active operations"

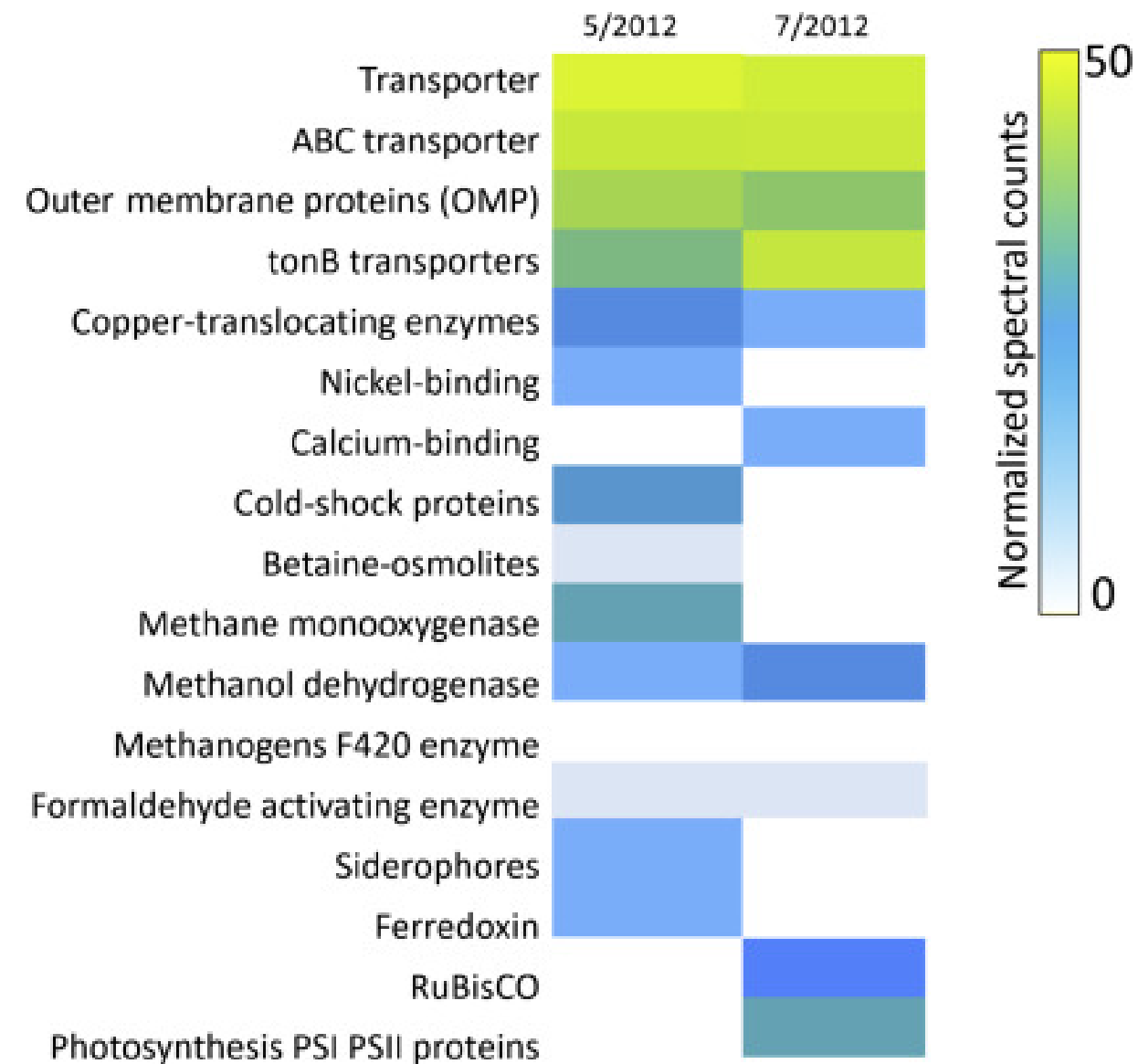


Reproduced from Ishii et al. (2013)

- Now we are studying all of the **mRNA** (transcriptome) molecules being produced by a **whole community** at a specific moment. In a nutshell, gene expression.
- This enables us to understand how a microbial community **responds** to their environment, new conditions or even disease.
- Consequently, we aim to answer the question "What is the community actually **doing right now?**".
- A new challenge arises: The signal-to-noise ratio. RNA is very unstable and degrades fast. You need to capture the "state of the system" under specific conditions.

Metaproteomics

The "Heat maps"



From Nunn, B.L., Gilmore, J., Timmins-Schiffman, E., Junge, K, 2018. Metaproteomics of Glacial Meltwater Reveals Taxonomic and Functional Difference Across Time and Strata

- Metaproteomics gives us a large-scale characterization of the **entire protein complement** of environmental microbiota at a given point in time.
- Metaproteomics provides a **direct view of microbial functional activity**, overcoming limitations of metagenomics (which reflects only metabolic potential) and metatranscriptomics (which does not account for translational regulation and can show weak mRNA–protein correlation).
- So now, we need to understand "What **physical** work is being completed?".

Thanks !!!!!